Microeconomics

One of the reasons that the government discourages and regulates monopolies is that
(A) producer surplus is lost and consumer surplus is gained. ✗
(B) monopoly prices ensure productive efficiency but cost society allocative efficiency. ✗
(C) monopoly firms do not engage in significant research and development. ✗
(D) consumer surplus is lost with higher prices and lower levels of output. ✓

Figure 3: Examples from the Microeconomics task.

Conceptual Physics

When you drop a ball from rest it accelerates downward at 9.8 m/s². If you instead throw it downward assuming no air resistance its acceleration immediately after leaving your hand is
(A) 9.8 m/s² ✓
(B) more than 9.8 m/s² ✗
(C) less than 9.8 m/s² ✗
(D) Cannot say unless the speed of throw is given. ✗

College Mathematics

In the complex z-plane, the set of points satisfying the equation $z^2 = |z|^2$ is a
(A) pair of points ✗
(B) circle ✗
(C) half-line ✗
(D) line ✓

Figure 4: Examples from the Conceptual Physics and College Mathematics STEM tasks.

real-world *text* understanding. Since models are pretrained on the Internet, this enables us to test how well they can extract useful knowledge from massive corpora. Future models that use this test could be single models or a mixture of experts model. To succeed at our test, future models should be well-rounded, possess extensive world knowledge, and develop expert-level problem solving ability. These properties make the test likely to be an enduring and informative goalpost.

3.1 Humanities

The humanities is a group of disciplines that make use of qualitative analysis and analytic methods rather than scientific empirical methods. Branches of the humanities include law, philosophy, history, and so on (Appendix B). Mastering these subjects requires a variety of skills. For example, legal understanding requires knowledge of how to apply rules and standards to complex scenarios, and also provide answers with stipulations and explanations. We illustrate this in Figure 2. Legal understanding is also necessary for understanding and following rules and regulations, a necessary capability to constrain open-world machine learning models. For philosophy, our questions cover concepts like logical fallacies, formal logic, and famous philosophical arguments. It also covers moral scenarios, including questions from the ETHICS dataset (Hendrycks et al., 2020) that test a model's understanding of normative statements through predicting widespread moral intuitions about diverse everyday scenarios. Finally, our history questions cover a wide range of time periods and geographical locations, including prehistory and other advanced subjects.

3.2 Social Science

Social science includes branches of knowledge that examine human behavior and society. Subject areas include economics, sociology, politics, geography, psychology, and so on. See Figure 3 for an example question. Our economics questions include microeconomics, macroeconomics, and econometrics, and cover different types of problems, including questions that require a mixture of world knowledge, qualitative reasoning, or quantitative reasoning. We also include important but more esoteric topics such as security studies in order to test the boundaries of what is experienced and learned during pretraining. Social science also includes psychology, a field that may be especially important for attaining a nuanced understanding of humans.

3.3 Science, Technology, Engineering, and Mathematics (STEM)

STEM subjects include physics, computer science, mathematics, and more. Two examples are shown in Figure 4. Conceptual physics tests understanding of simple physics principles and may be thought

Professional Medicine

A 33-year-old man undergoes a radical thyroidectomy for thyroid cancer. During the operation, moderate hemorrhaging requires ligation of several vessels in the left side of the neck. Postoperatively, serum studies show a calcium concentration of 7.5 mg/dL, albumin concentration of 4 g/dL, and parathyroid hormone concentration of 200 pg/mL. Damage to which of the following vessels caused the findings in this patient?

(A) Branch of the costocervical trunk ✗
(B) Branch of the external carotid artery ✗
(C) Branch of the thyrocervical trunk ✓
(D) Tributary of the internal jugular vein ✗

Figure 5: A question from the Professional Medicine task.

of as a harder version of the physical commonsense benchmark Physical IQA (Bisk et al., 2019). We also test mathematical problem solving ability at various levels of difficulty, from the elementary to the college level. College mathematics questions, like those found on the GRE mathematics subject test, often require chains of reasoning and abstract knowledge. To encode mathematics expressions, we use LaTeX or symbols such as * and ˆ for multiplication and exponentiation respectively. STEM subjects require knowledge of empirical methods, fluid intelligence, and procedural knowledge.

3.4 Other

There is a long tail of subjects that either do not neatly fit into any of the three preceding categories or for which there are not thousands of freely available questions. We put these subjects into Other. This section includes the Professional Medicine task, which has difficult questions that require humans many years of study to master. An example is depicted in Figure 5. This section also contains business topics like finance, accounting, and marketing, as well as knowledge of global facts. The latter includes statistics about poverty in different countries over time, which may be necessary for having an accurate model of the world internationally.

4 Experiments

4.1 Setup

Assessment and Models. To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks. We evaluate GPT-3 (Brown et al., 2020) and UnifiedQA (Khashabi et al., 2020). For GPT-3 we use the OpenAI API, which provides access to four model variants, "Ada," "Babbage," "Curie," and "Davinci," which we refer to as "Small" (2.7 billion parameters), "Medium" (6.7 billion), "Large" (13 billion) and "X-Large" (175 billion). UnifiedQA uses the T5 (Raffel et al., 2019) text-to-text backbone and is fine-tuned on previously proposed question answering datasets (Lai et al., 2017), where the prediction is the class with the highest token overlap with UnifiedQA's text output. Since UnifiedQA is fine-tuned on other datasets, we evaluate it without any further tuning to assess its transfer accuracy. We also fine-tune RoBERTa-base, ALBERT-xxlarge, and GPT-2 on UnifiedQA training data and our dev+val set. We primarily focus on UnifiedQA and GPT-3 in the rest of this document, but additional discussion of RoBERTa, ALBERT, and GPT-2 is in Appendix A.

Model	Humanities	Social Science	STEM	Other	Average
Random Baseline	25.0	25.0	25.0	25.0	25.0
RoBERTa	27.9	28.8	27.0	27.7	27.9
ALBERT	27.2	25.7	27.7	27.9	27.1
GPT-2	32.8	33.3	30.2	33.1	32.4
UnifiedQA	45.6	56.6	40.2	54.6	48.9
GPT-3 Small (few-shot)	24.4	30.9	26.0	24.1	25.9
GPT-3 Medium (few-shot)	26.1	21.6	25.6	25.5	24.9
GPT-3 Large (few-shot)	27.1	25.6	24.3	26.5	26.0
GPT-3 X-Large (few-shot)	40.8	50.4	36.7	48.8	43.9

Table 1: Average weighted accuracy for each model on all four broad disciplines. All values are percentages. Some models proposed in the past few months can move several percent points beyond random chance. GPT-3 uses few-shot learning and UnifiedQA is tested under distribution shift.

Few-Shot Prompt. We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject.

4.2 Results

Model Size and Accuracy. We compare the few-shot accuracy of each GPT-3 size in Table 1. We find that the three smaller GPT-3 models have near random accuracy (around 25%). In contrast, we find that the X-Large 175 billion parameter GPT-3 model performs substantially better than random, with an accuracy of 43.9%. We also find qualitatively similar results in the zero-shot setting. While the smaller models have around 25% zero-shot accuracy, Figure 10 in Appendix A shows that the largest GPT-3 model has a much higher zero-shot accuracy of about 37.7%. Brown et al. (2020) also observe that larger GPT-3 models perform better, though progress tends to be steadier. In Figure 1b we show that non-random accuracy on the multitask test emerged with recent large few-shot models compared to datasets that assess commonsense and linguistic understanding.

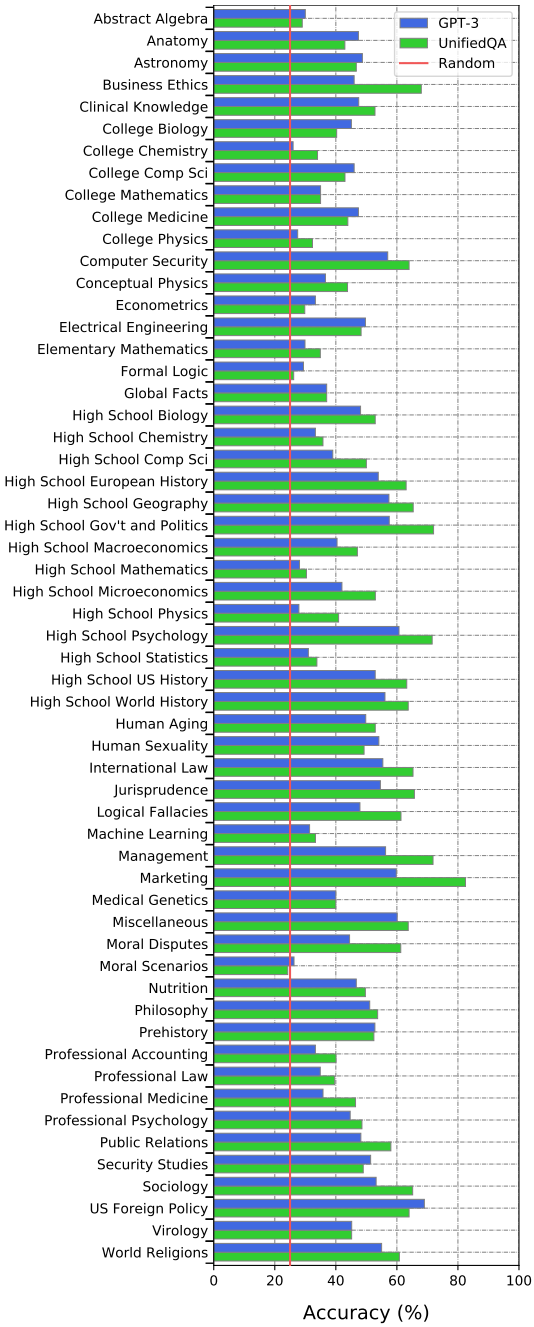

Figure 6: GPT-3 (few-shot) and UnifiedQA results.

To test the usefulness of fine-tuning instead of few-shot learning, we also evaluate UnifiedQA models. UnifiedQA has the advantage of being fine-tuned on other question answering datasets, unlike GPT-3. We assess UnifiedQA by evaluating its transfer performance without any additional fine-tuning. The largest UnifiedQA model we test has 11 billion parameters, which is slightly smaller than GPT-3 Large. Nevertheless, we show in Table 1 that it attains 48.9% accuracy. This performs better than the few-shot GPT-3 X-Large model, despite UnifiedQA have an order of magnitude fewer parameters. We also find that even the smallest UnifiedQA variant, with just 60 million parameters, has approximately 29.3% accuracy. These results suggest that while model size is a key component for achieving strong performance, fine-tuning also helps.

Comparing Disciplines. Using our test, we discover that GPT-3 and UnifiedQA have lopsided performance and several substantial knowledge gaps. Figure 6 shows the accuracy of GPT-3 (few-shot) and UnifiedQA for all 57 tasks. It shows the both models are below expert-level performance for all tasks, with GPT-3's accuracy ranging from 69% for US Foreign Policy to 26% for College Chemistry. UnifiedQA does best on marketing, with an accuracy of 82.5%.

Overall, models do poorly on highly procedural problems. Figure 6 shows that calculation-heavy STEM subjects tend to have low accuracy compared to verbal subjects. For GPT-3, 9 out of the 10